

Rye whole grain and bran intake compared with refined wheat decreases urinary C-peptide, plasma insulin, and prostate specific antigen in men with ...

Rye whole grain and bran intake has shown beneficial effects on prostate cancer progression in animal models, including lower tumor take rates, smaller tumor volumes, and reduced prostate specific antigen (PSA) concentrations. A human pilot study showed increased apoptosis after consumption of rye bran bread. In this study, we investigated the effect of high intake of rye whole grain and bran on prostate cancer progression as assessed by PSA concentration in men diagnosed with prostate cancer. Seventeen participants were provided with 485 g rye whole grain and bran products (RP) or refined wheat products with added cellulose (WP), corresponding to ~50% of daily energy intake, in a randomized controlled, crossover design. Blood samples were taken from fasting men before and after 2, 4, and 6 wk of treatment and 24-h urine samples were collected before the first intervention period and after treatment. Plasma total PSA concentrations were lower after treatment with RP compared with WP, with a mean treatment effect of -14% ($P = 0.04$). Additionally, fasting plasma insulin and 24-h urinary C-peptide excretion were lower after treatment with RP compared with WP ($P < 0.01$ and $P = 0.01$, respectively). Daily excretion of 5 lignans was higher after the RP treatment than after the WP treatment ($P < 0.001$). We conclude that whole grain and bran from rye resulted in significantly lower plasma PSA compared with a cellulose-supplemented refined wheat diet in patients with prostate cancer. The effect may be related to inhibition of prostate cancer progression caused by decreased exposure to insulin, as indicated by plasma insulin and urinary C-peptide excretion.